SELECTED ENGINEERING WORK

Christopher Yanes

MECHANICAL ENGINEERING PORTFOLIO

CAD modeling. Technical drawings. Manufacturing process.



CNC fabrication



CAD reconstruction

Mechanical engineering student at California Baptist University, expecting to graduate in April 2028. Seeking a summer internship in mechanical engineering, CAD/design, or manufacturing.

TOOLS & CREDENTIAL

SOLIDWORKS / AutoCAD / Mastercam / Haas CNC milling
SOLIDWORKS CAD Design Associate (CSWA), issued May 2025

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LinkedIn profile

Selected project work | CAD, fabrication, and simulation studies

SELECTED WORK / 01

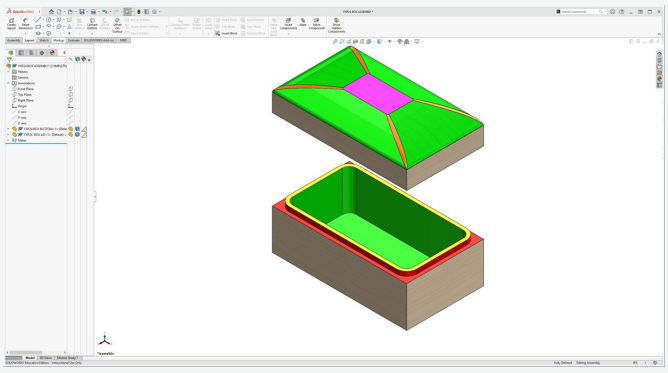
CNC wood box & engraved lid

SOLIDWORKS / MASTERCAM / CNC MILLING / LASER ENGRAVING

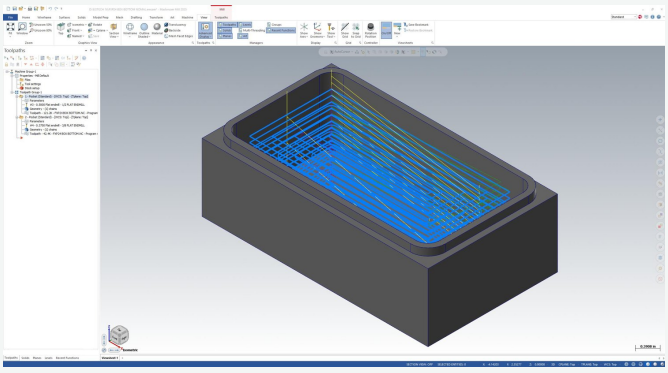
Modeled and manufactured a wooden enclosure with an interior cavity and separate lid. The project connects stock preparation, CAD/CAM planning, machining, measurement, and an engraved finish.



Finished enclosure and lid, showing the machined cavity and mating geometry.



01 / Model the body and lid as separate components.



02 / Program the cavity machining passes in Mastercam.

WHAT THIS WORK SHOWS

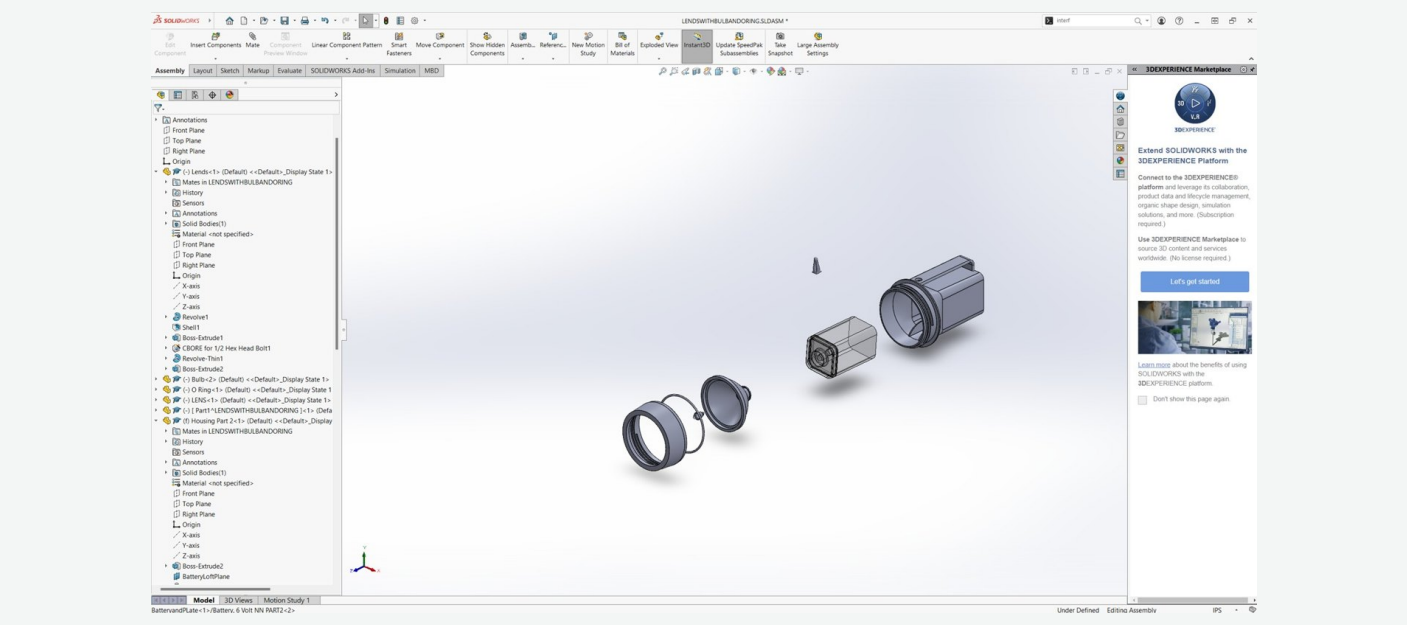
A physical finished part supported by a documented CAD-to-CAM workflow. Additional gallery images show marked stock, caliper measurement, machining, and lid engraving.

SELECTED WORK / 02

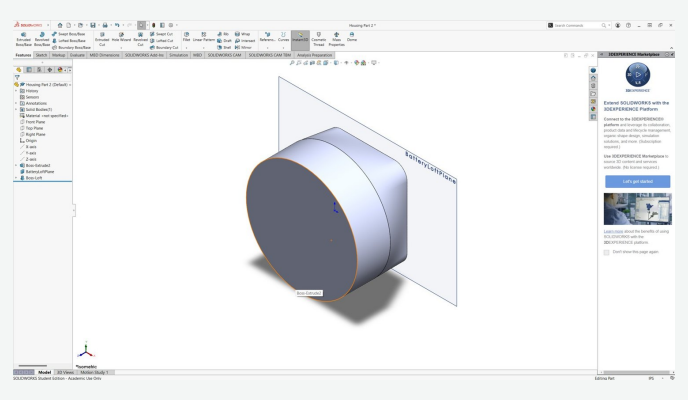
Flashlight housing & assembly

SOLIDWORKS / FEATURE MODELING / ASSEMBLY DOCUMENTATION

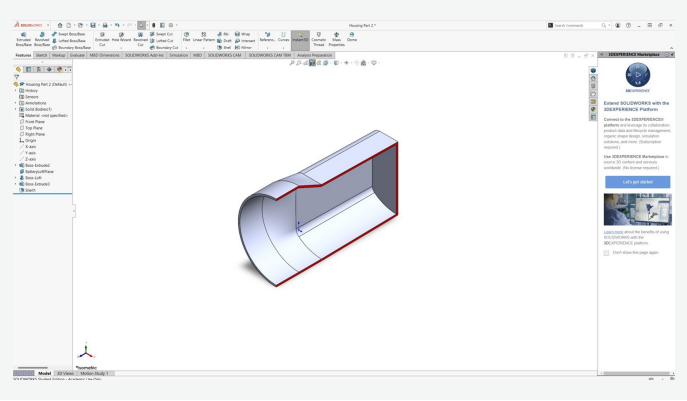
Built a flashlight housing and component assembly from reference geometry. The model combines lofted surfaces, a hollow shell, ribs, threads, and fitted components, documented through section and exploded views.



Exploded CAD assembly showing the relationship between the housing, battery, lens, and cap.



01 / Develop the housing transition with a lofted feature.



02 / Create the hollow interior with a shell feature.

WHAT THIS WORK SHOWS

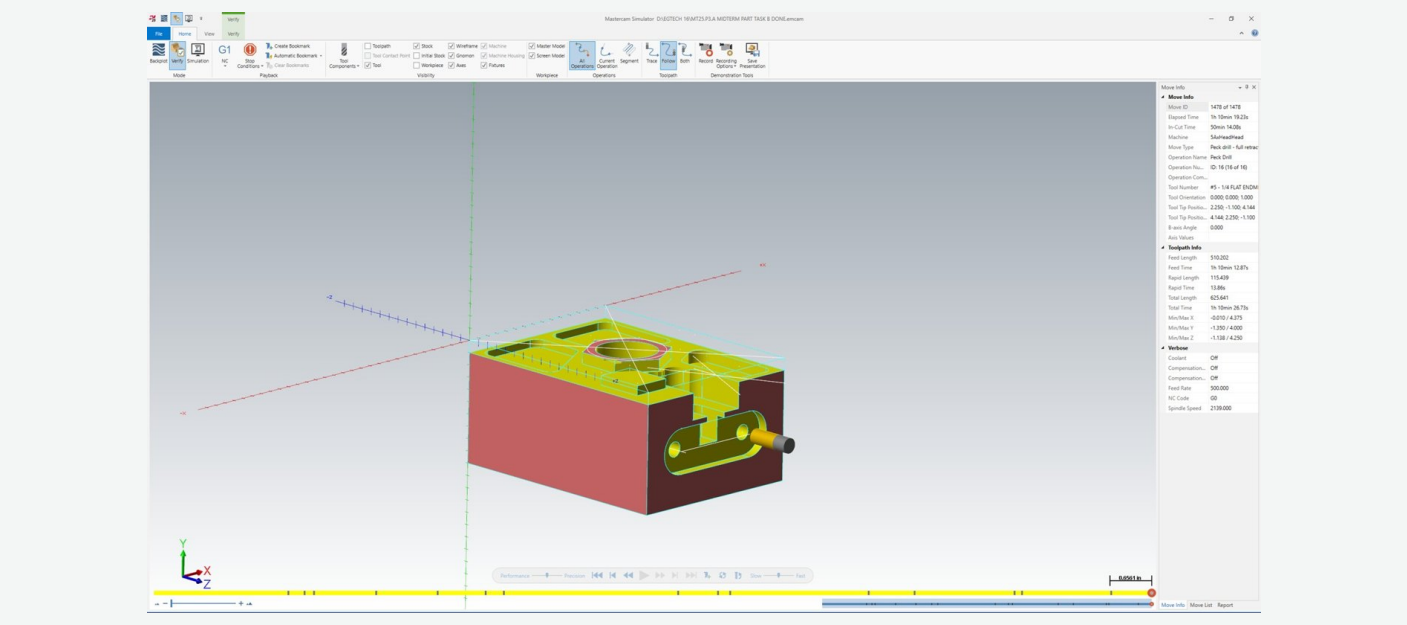
Feature-based part construction and communication of assembly structure. This is a CAD modeling study; the gallery includes section views and interference inspection for further review.

SELECTED WORK / 03

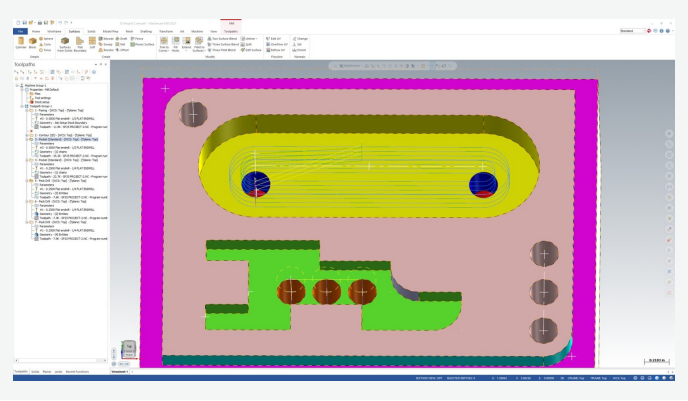
Multi-setup CNC programming

MASTERCAM / OPERATION SEQUENCING / TOOLPATH VERIFICATION

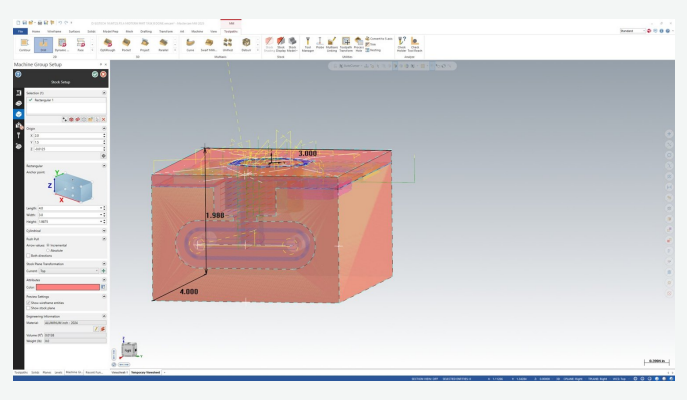
Created machining plans for reference parts with pockets, bosses, bores, and side features. Organized stock and toolpath operations around part orientation, with multiple setups and simulated material removal.



Verification view of simulated machining for a part with features on multiple faces.



01 / Plan passes for the pocket and surrounding features.



02 / Reorient the part for the next machining setup.

WHAT THIS WORK SHOWS

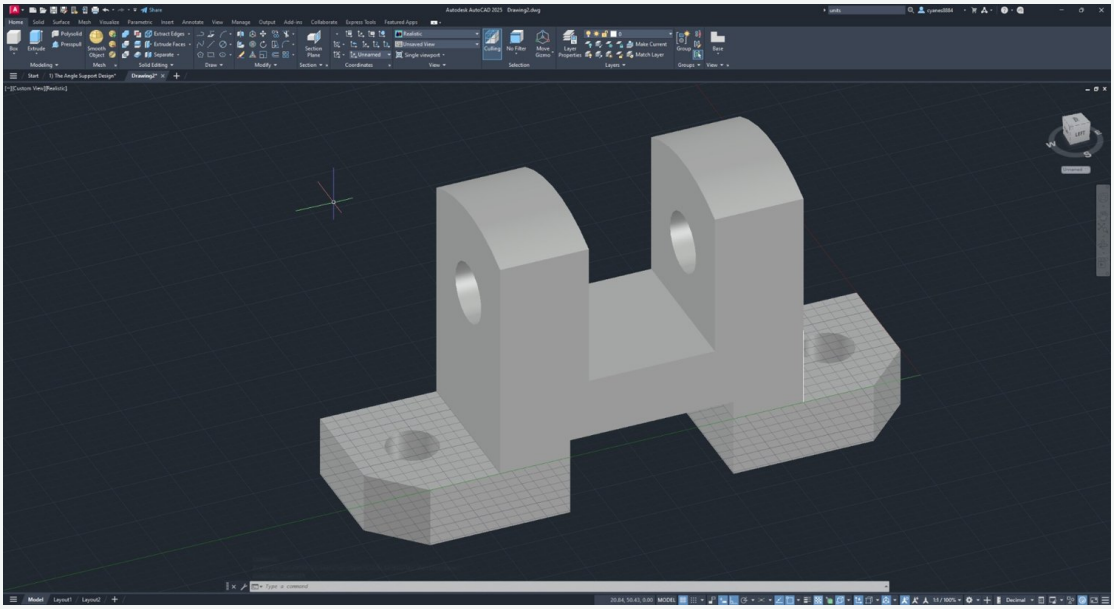
CAM planning across setups, operation sequencing, and simulation review. Additional portfolio studies include contoured plates and curved-surface toolpaths. Shown results are simulations.

SELECTED WORK / 04

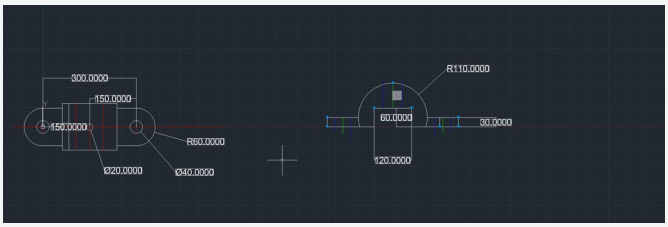
Mechanical profiles & brackets

AUTOCAD / 2D DRAFTING / 3D MODELING

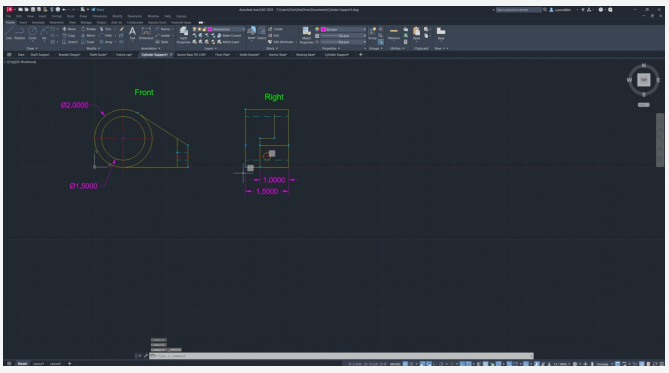
Created mechanical profiles, dimensioned orthographic views, and solid bracket models. The work connects geometric construction in two dimensions with clear communication of three-dimensional part features.



Shaded bracket model with mounting features and upright supports.



01 / Communicate geometry with dimensioned fixture views.



02 / Use front and side views to describe a support part.

WHAT THIS WORK SHOWS

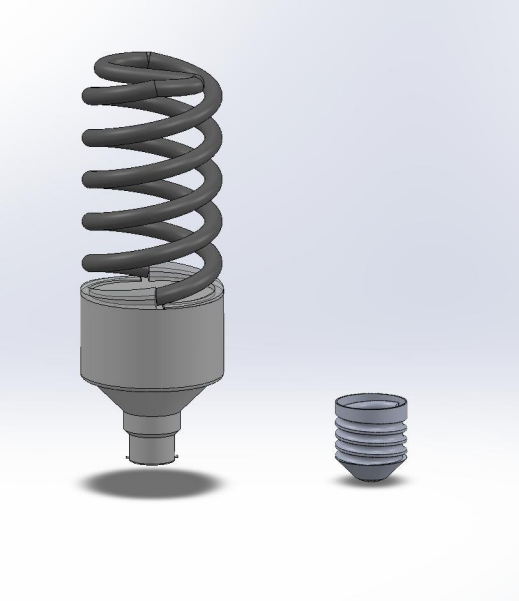
Translating part geometry into profiles, multiple views, and solid models. The full portfolio includes further bracket examples and jig, block, and section-drawing studies.

SELECTED WORK / 05

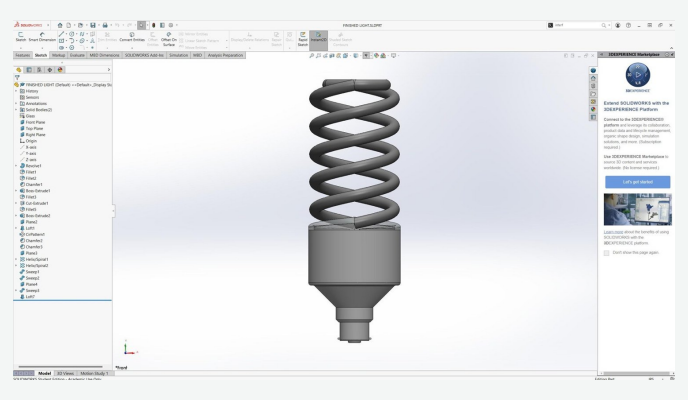
Spiral light bulb reconstruction

SOLIDWORKS / REVERSE ENGINEERING / DRAWING DOCUMENTATION

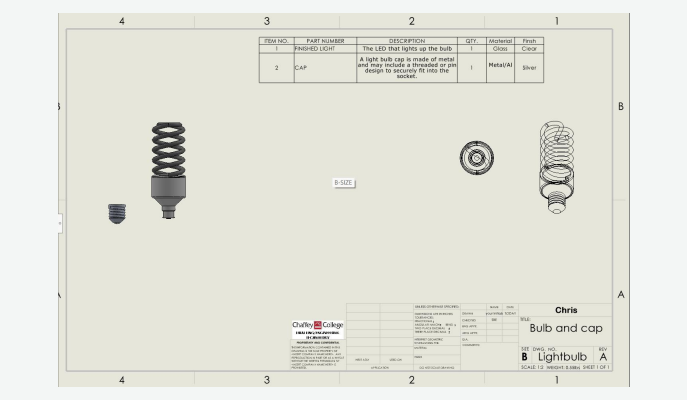
Recreated the external form of a spiral light bulb using physical references. Developed the body and screw-cap models, assembly and exploded views, and drawing sheets to communicate the component structure.



Exploded view showing the reconstructed spiral body and separate screw cap.



01 / Reconstruct the spiral form and housing in CAD.



02 / Document the model using drawing views and a parts list.

WHAT THIS WORK SHOWS

Observation, component modeling, and technical presentation of an existing product. The project studies external geometry and assembly structure; it does not include electrical design or testing.